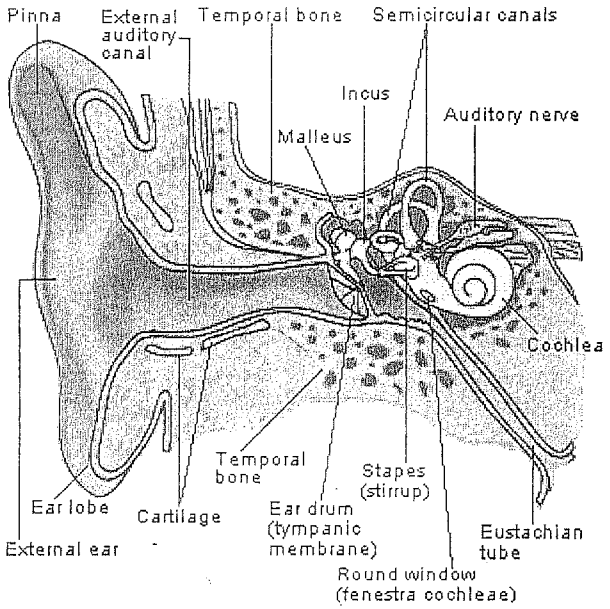
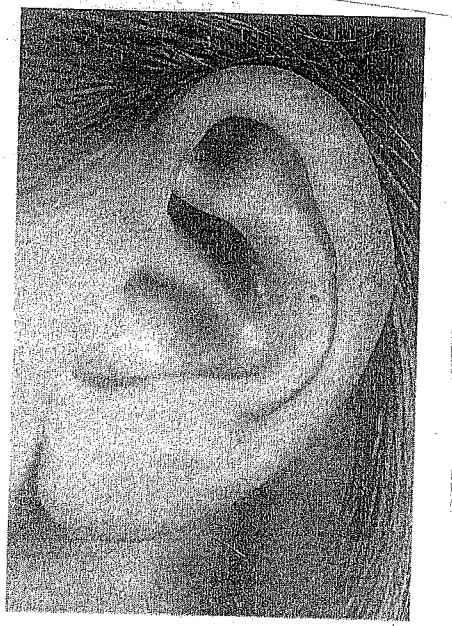
Ear

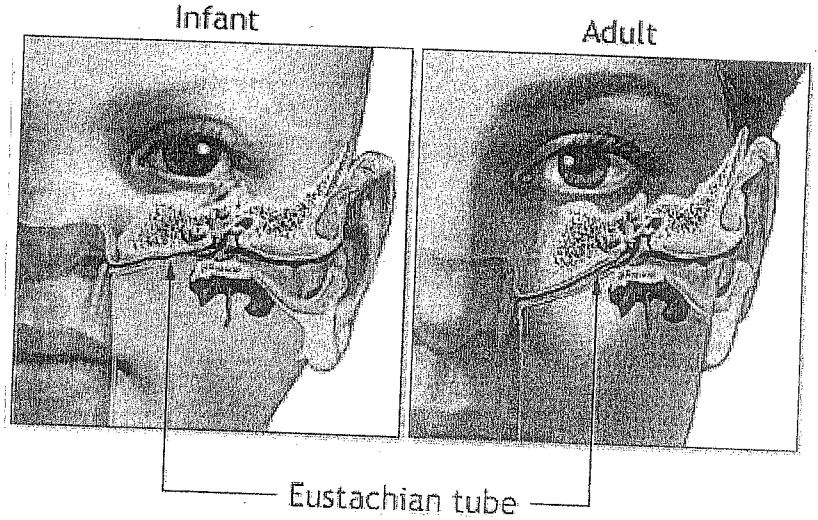
Joe is impressed by the computer his company bought not long ago. It will perform seeming miracles, but to me it is still as crude as a concrete mixer. Perhaps I am prejudiced, for I am a triumph of miniaturization. Nowhere in his body is so much crammed into so small a space as in me. I have enough electrical circuits to provide phone service for a good-sized city. I am also a kind of automatic pilot keeping Joe from toppling over.

I am Joe's right ear, and I do all this in a space not much larger than a hazelnut! Joe considers his eyes his most important sensory organs. Yet, without my partner and me, he would be doomed - to solitary sonic confinement--far more emotionally disabling than blindness.

Joe thinks of me simply as that flap of tissue on the side of his head. This outer ear, however, is nothing but a sound-gathering trumpet. From it a three-centimeter canal runs obliquely to the ear-drum, twisting to protect my delicate inner components and warming air to keep things cozy. In this canal, a profusion of hairs and 4000 wax glands act as a flypaper trap for insects, dust and other potential irritants. Further, the wax guards against infection, particularly when Joe swims in dirty water. (He can wash away unsightly wax, but I wish he would not pick at the rest--he could harm my eardrum,







and I will shed excess wax naturally.)

My eardrum, a tough, tightly stretched membrane less than two centimeters across, is where the intricate business of hearing starts. Sound-bearing airwaves strike it--like a stick beating a drum. Even faint vibrations from a whisper can push it inward--but ever so little, perhaps only a billionth of a centimeter. This minute displacement is then changed, in an awe-inspiring chain of events not yet entirely understood, into meaningful sound for Joe.

To see how, step through my drumhead to Joe's bean-size middle ear. Here are hinged together three tiny bones called the anvil, hammer and stirrup (also known as the stapes) because they vaguely resemble those things. It is their job to step up the tiny movements of my drum, amplifying them 22 times and passing them on to my inner ear via oval window attached to the stirrup. My inner ear--the real organ of hearing--resides in a fortress like cavern hollowed out of the body's hardest bone and filled wit watery fluid. Its major hearing component is the snail-shaped cochlea, whose twisting interior is studded with thousands of microscopic hair like nerve cells, each one tuned to a particular vibration. When the middle ear's stirrup "knocks" on the oval window leading to the inner ear, this fluid is set vibrating. If middle C has been sounded, say, then cochlea's middle C's hair cell vibrates, waving in the lymph fluid like seaweed in a tidal current.

The waving produces a wisp of electricity that feeds into my auditory nerve (only the diameter of a pencil lead, this nerve contains more than 30,000 circuits!), which m turn leads to Joe's brain, two centimeters away. My cochlea may feed in thousands of electrical messages, with Joe's left ear doing the same. It is the brain's job to unscramble these data and convert them into meaningful sound. Thus, Joe hears with me, but in his brain.

So far I have talked only of sound conducted by airwaves. Joe can also hear by bone conduction. When Joe speaks, part of the sound leaves his mouth and strikes my drum, but another part travels directly to my inner-ear fluid via the jawbones. Thus, what Joe hears is quite different from what a listener hears. That is why he has trouble recognizing his own voice on a tape recorder. It is also why Joe thinks he is making quite a racket when he eats celery.

But hearing is only part of the story of my miraculous inner ear. Above the cochlea I have three minute, fluid-filled semicircular canals. These loops of tubing are Joe's organ of balance. One detects up-and-down motion, another forward motion, the third lateral motion. If Joe starts to fall, fluid in one of my canals is displaced. Hair cells there detects this and inform Joe's brain which orders muscles to tighten to keep him upright.

As a youngster, Joe sometimes liked to be whirled by another boy until he was staggering. What happened was this: fluid

in the canals was being displaced so rapidly that the brain was getting more messages than it could handle, and Joe lost all muscular control, Let disorderly displacement of fluid continue too long, as in a tossing boat, and I begin to involve other organs. Joe breaks into a sweat, and motion sickness is apt to follow.

Joe's hearing started declining almost the moment he was born. It is now going down each year as my tissues lose elasticity, hair cells degenerate and calcium deposits invade critical spots. When Joe was a baby, he had a hearing range of 16 to 30,000 cycles per second (vibrations). (If it had gone much below 16 he could have heard the vibrations of his own body. As a matter of fact, Joe can hear his body vibrations. Let him stopper his ears with his finger: the low rumble he hears comes from tensed finger and arm muscles.) By the time he reached his teens, the upper limit of his hearing had dropped to 20,000 cycles. Now he hears nothing above 8000, and, if he reaches the age of 80, that will be down to about 4000. He will then hear conversation reasonably well in a quiet place, but may have difficulty in a noisy area. He will hear low tones better than high tones.

He also has a decibel loss. Decibels measure sound intensity at any particular frequency. Thus, a whisper from four feet away in a quiet room ranks at about 30 decibels, normal conversation about 60,a rock band 120, and a shotgun 140. (But this does not mean that a rock band is only twice as loud as normal conversation. A jump of twenty points on the tricky decibel scale means a hundredfold increase of intensity.) Right now Joe has a 40-decibel loss; his hearing is quite serviceable, but he is beginning to ask people to repeat words.

With any structure as complex as mine, a great deal can go wrong. Drum punctures are frequent-fortunately, most such punctures heal themselves or can be repaired by an operation. Tinnitus, or ringing in the ears, is another source of trouble. This can be caused by almost anything: drugs (some antibiotics, alcohol), fever, circulation changes, and tumors on my acoustic nerve. Once the causative factor is tracked down and eliminated, I sometimes cease my racket.

Middle-ear infections are another source of trouble--and, before antibiotics, often culminated in hearing loss. The eustachian tube, leading from the middle ear to Joe's throat, is the culprit. The throat is, microbially speaking, a very dirty place, and the eustachian tube offers microbes easy access to the middle ear. When he has a cold, Joe would be wise not to blow his nose too hard-- it forces throat pollution into me.

Sometimes an overgrowth of bone is apt to freeze motion of the bones in my middle ear. Once motion stops, hearing is impaired. This is conduction deafness. Joe has the beginnings of it, but chances of its progressing to really serious deafness are only about one in ten. If this happens, Joe has two options: a hearing aid or surgery. The operation (which

has an 80 percent chance of success) replaces my stirrup bone with a tiny filament of stainless steel. Motion of the bones resumes, and Joe can hear again.

Perhaps the biggest thing Joe should be worrying about right now is noise pollution. Joe knows that workers in noisy trades may develop hearing difficulty, and that today's rock musicians will probably be wearing hearing aids a few years hence. But he thinks he can adjust to today's shrill racket. He cannot. When excessively loud, low-pitched sounds strike my drum, I have muscles to tighten it; otherwise, I take what comes. This was fine for Joe's ancestors. Thunder or the roar of a lion were the loudest sounds around, and these were low-pitched. It is the new high-pitched sounds-such as the whine of jets, the rat-a-tat-tat of riveting machines-that wreck me.

Sustained loud noise can wreck internal organs of a mouse and eventually kill it. If such an experiment were ever tried on Joe, I can guess the result. What can Joe do about it? He could speak out against senseless noise, seek quiet in home and office and cover his ears when he goes hunting--the repeated burst of a shotgun can really wreck me. He could stop smoking, or at least cut down. Nicotine, coffee, too constricts arteries in my all-important inner ear, reducing the nourishment my inner ear needs.

Joe has his eyes examined regularly, and I would like the same attention. If Joe only knew how limited and lonely the world of silence is, he would take all possible steps to preserve my partner and me.

Q1. What is the role of ear wax?
Q2. What is the range of human hearing in Hertz?
Q3. What is the role of the cochlea?
Q4. What is the role of the middle ear bones/ossicles?
Q5. Why do we suffer hearing loss and what activities are likely to do this?
Q6. What are middle ear infections and why are they so damaging?
Q7. What is noise pollution?
Q8. Can you explain the difference between volume and pitch?